STATSEI14 · PRACTICAL TUTORIAL

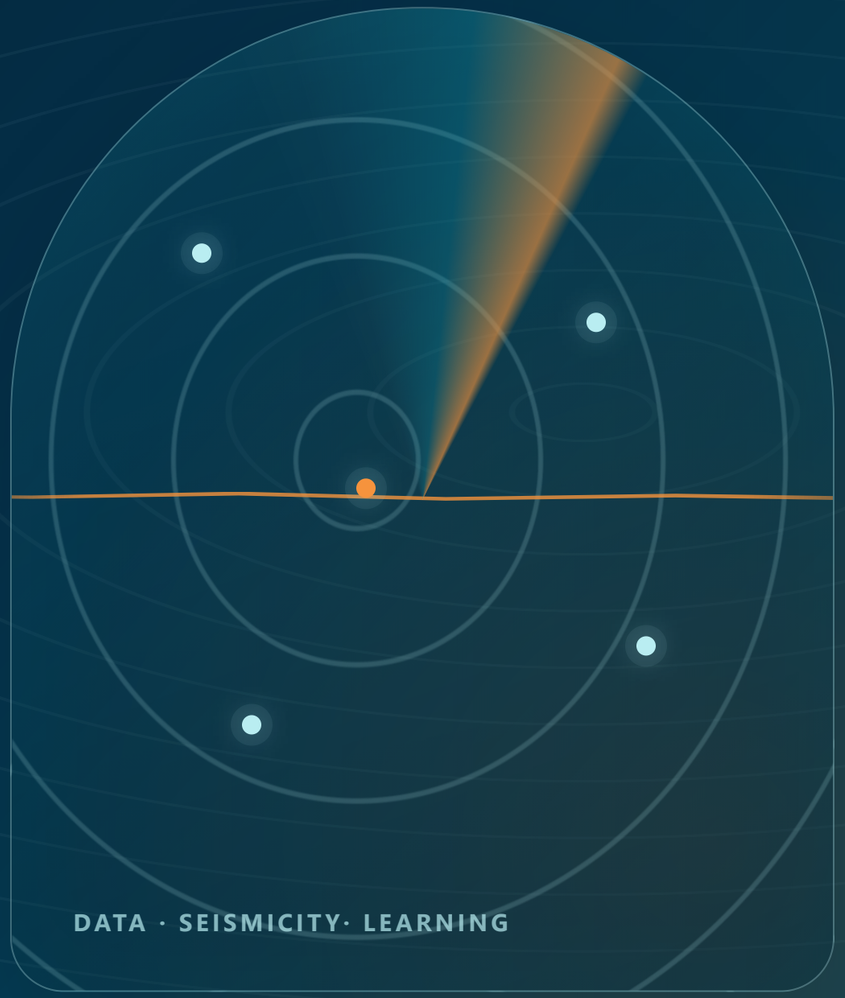
Deep Learning Meets ETAS

Practical Hybrid Workflows for Seismic Forecasting

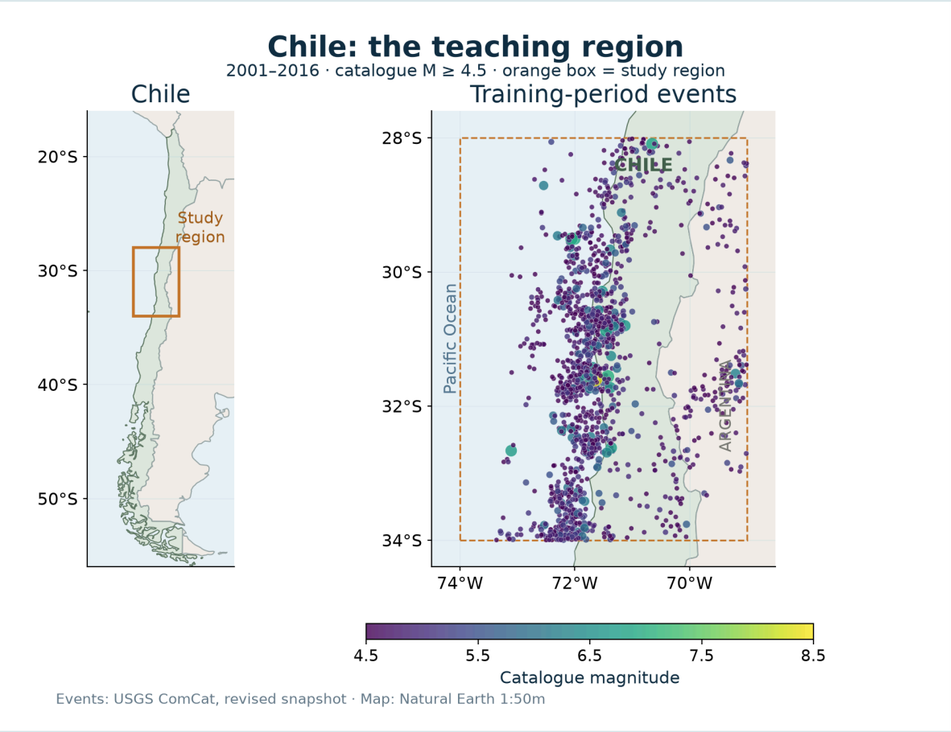
Francisco Plaza-Vega
Universidad de Santiago de Chile

13 October 2026 · Chile

> Seismic learning tasks · Research cases · Guided modelling and discussion



What can we learn from seismicity?



Signals reveal recorded events.

Catalogues describe event histories.

Models turn histories into questions about what may follow.

> What information do we represent, and what do we ask a model to learn?

Map: frozen USGS training catalogue and Natural Earth. Illustrates today's regional setting.

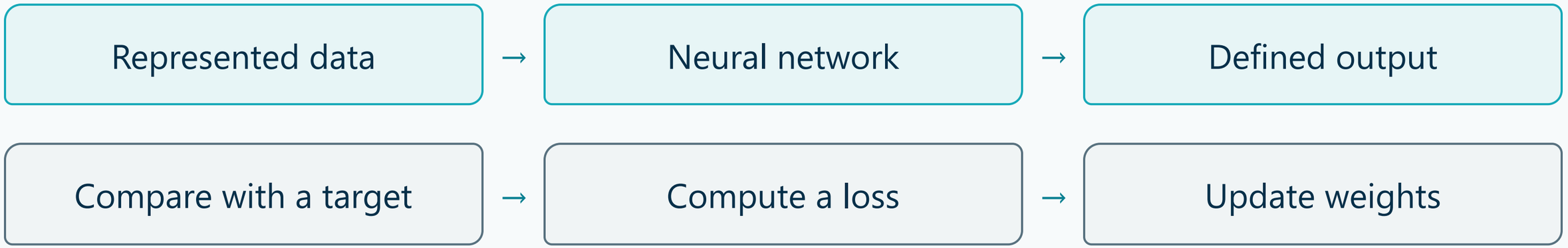
Deep learning in seismology: inputs and targets

Example	Input	Learning task
ConvNetQuake · 2018	Recorded waveforms	Detect events and classify their source region
RECAST · 2023	Past event times and magnitudes	Estimate the next-event time distribution
Delgado et al. · 2025	Features of catalogued events	Classify foreshock, mainshock and aftershock labels
Farfán et al. · 2025	Sequences of daily spatial grids	Forecast gridded seismic activity

> Today’s practical: 30 days of history and a seven-day occurrence probability.

Perol et al. · Dascher-Cousineau et al. · Delgado et al. · Farfán et al.

A network learns a mapping from examples

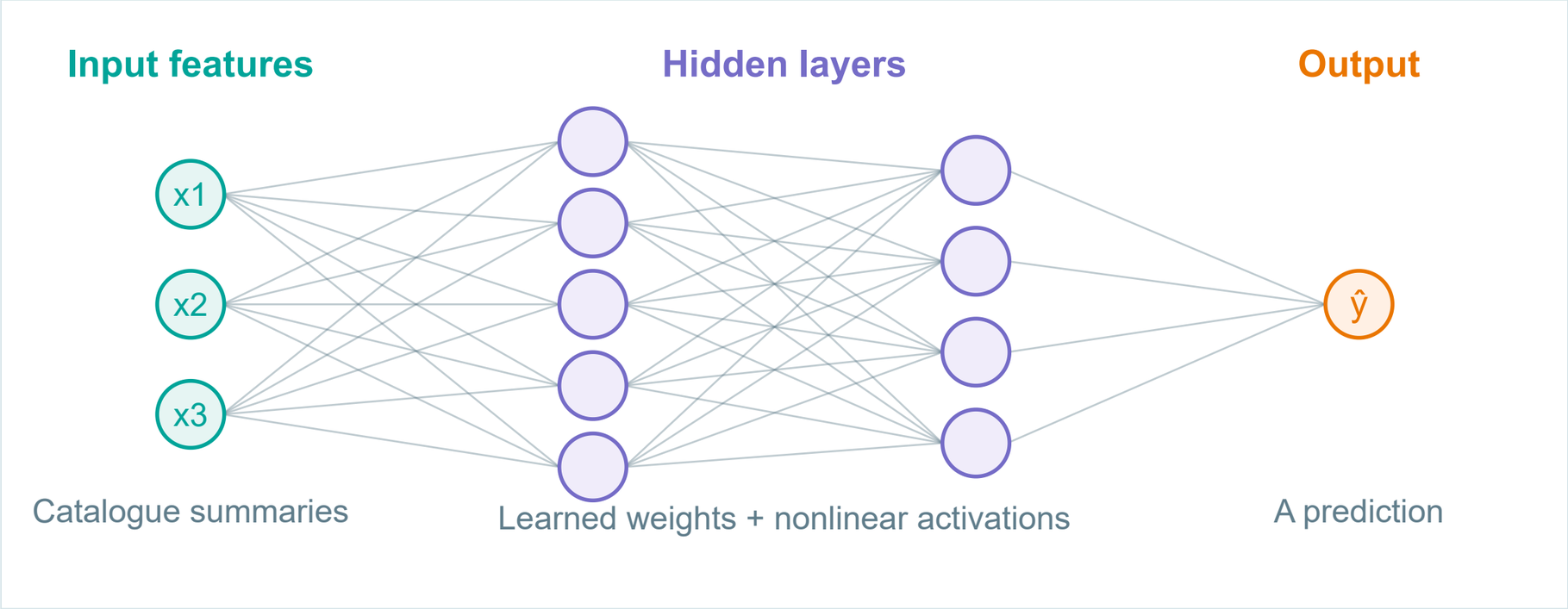


Architecture determines how information is combined.

Training adjusts weights to reduce the loss.

> The output and the loss must match the scientific question.

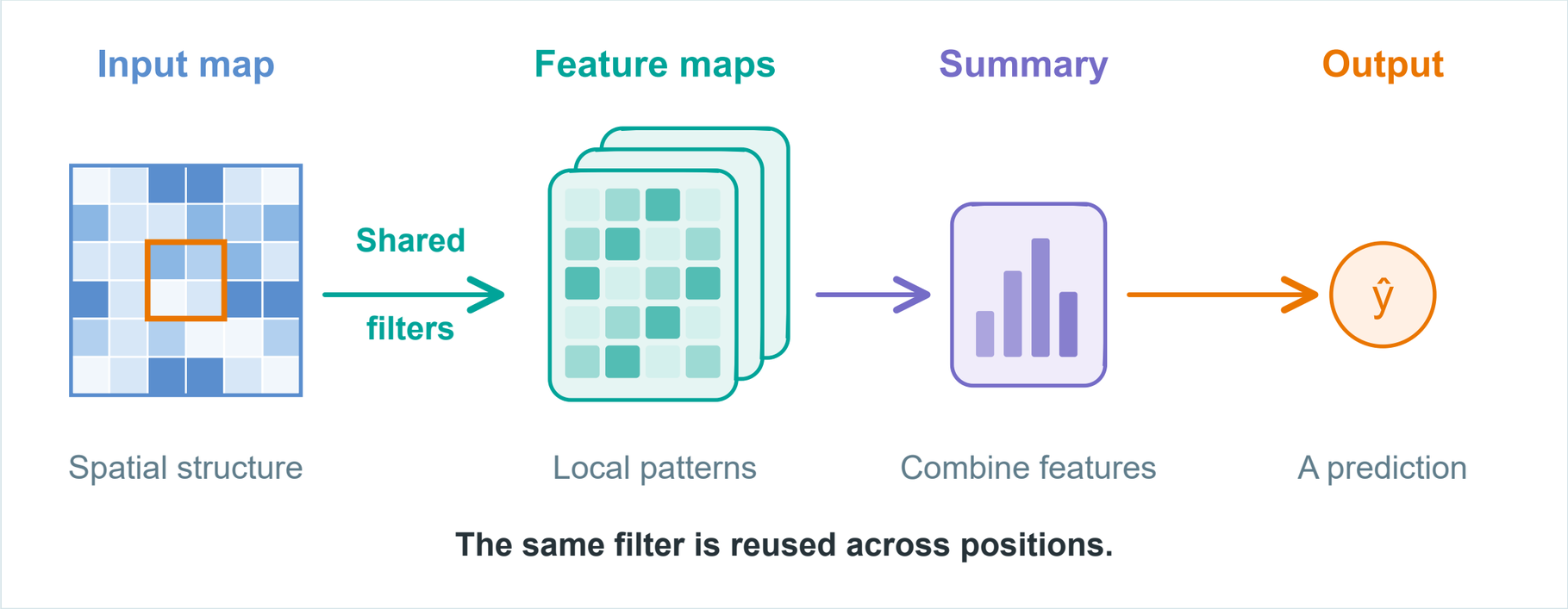
MLP: combine measured features



A multilayer perceptron learns nonlinear combinations of a feature vector.

Optional extension: the MLP uses 90 values at fixed lag positions. Diagram adapted from the author's DL course, Unit 2.

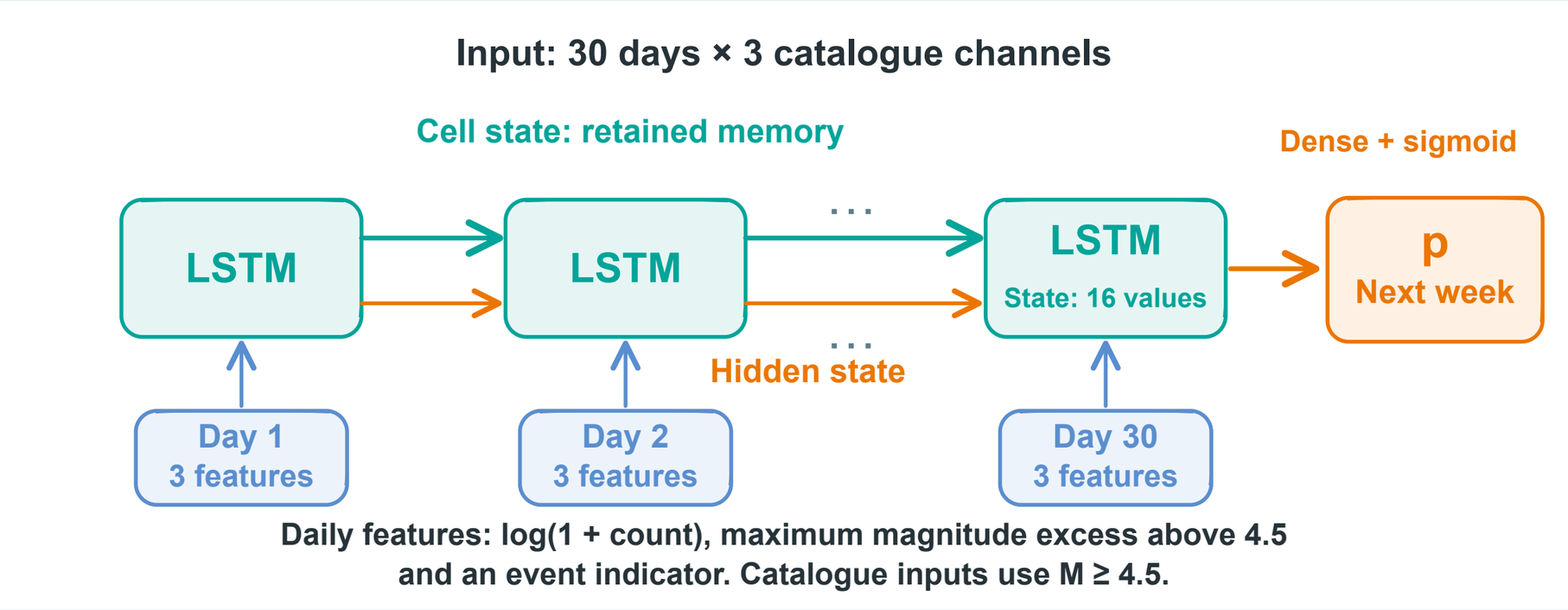
CNN: learn local patterns with shared filters



The same filter examines different neighborhoods.

Optional extension: CNN1D shares filters across days. Diagram adapted from the author's DL course, Unit 3.

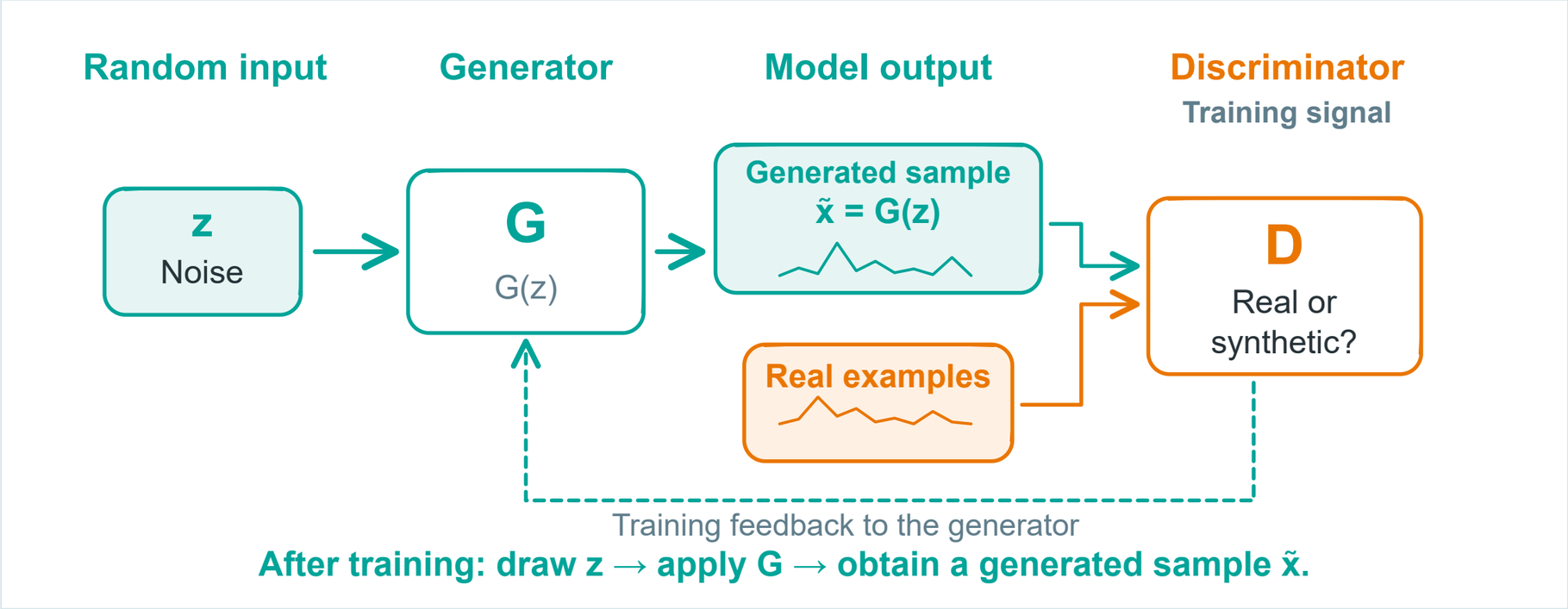
LSTM: carry information through a sequence



Today's example: 30 × 3 input → state (16) → Dense + sigmoid → weekly probability

Output: probability of at least one catalogue M ≥ 5 event in the next seven days.
Adapted from the author's DL course, Unit 4.

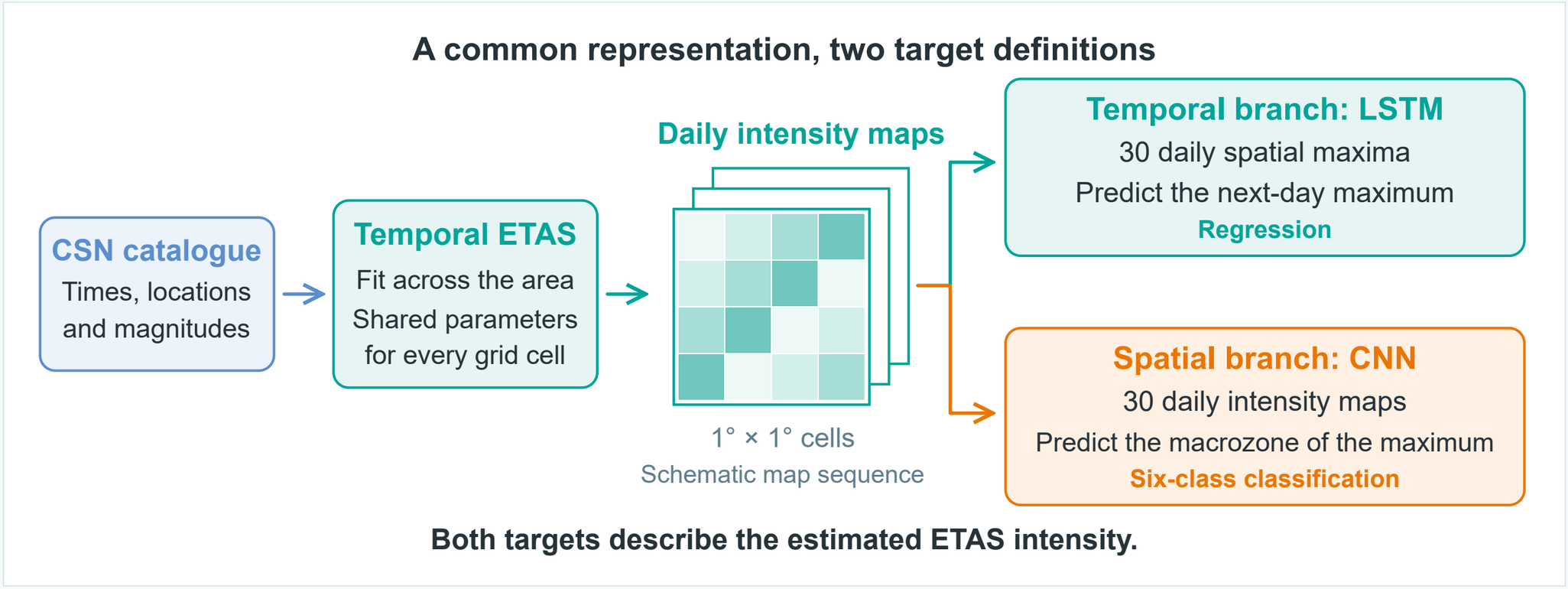
Generative learning: produce new samples



> The output of G is a new data sample, such as a synthetic sequence.

Later case: mainshock conditions + noise → a possible aftershock sequence.
GAN schematic adapted from the author's DL course, Unit 5.

ETAS intensity maps support two learning tasks



Nicolis, Plaza & Salas · Spatial Statistics 42 (2021), 100442 · CSN Chile, 2000–2017

ETAS intensity describes a conditional event rate

$$\lambda(t \mid \mathcal{H}_t) = \mu + \sum_{t_i < t} \text{contribution from event } i$$

Background + magnitude-dependent contributions that decay with time

Published construction: common ETAS parameters applied across 1° × 1° cells produce a sequence of intensity maps.

> A map value is an estimated occurrence rate, not an earthquake magnitude.

LSTM regression of ETAS-derived intensity

30 daily spatial maxima → LSTM → Next-day maximum intensity

One daily value: the maximum estimated ETAS intensity across the map.

Input: 30 consecutive values. **Target:** the following day's value.

> This LSTM predicts a continuous intensity summary.

Nicolis, Plaza & Salas (2021) · Regression of a model-derived target

CNN classification of six macrozones

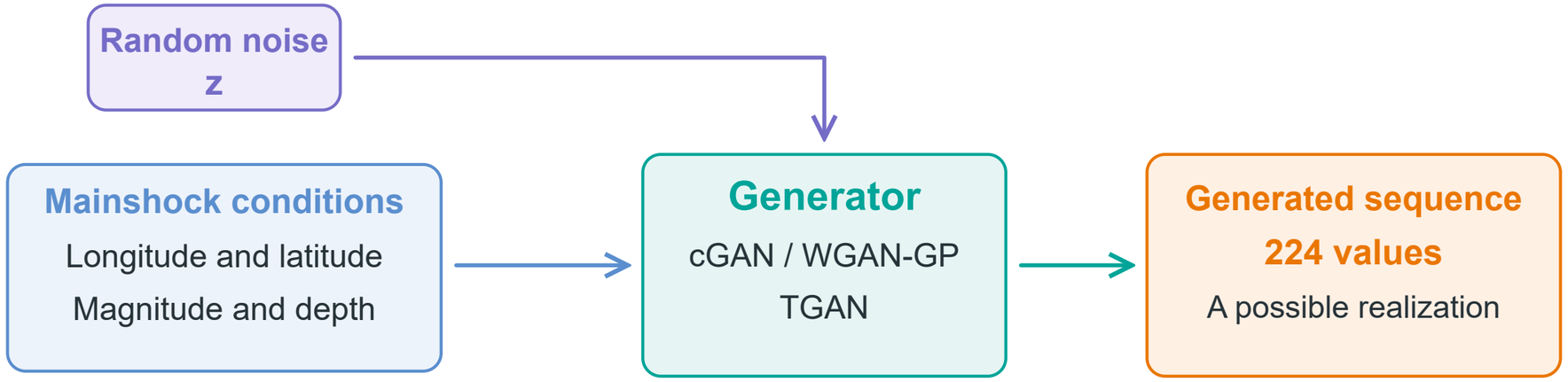
30 daily ETAS maps → CNN → Six macrozone probabilities

Which latitudinal macrozone contains the next day’s maximum estimated intensity?

> One of six location classes describes the maximum of the ETAS map.

Nicolis, Plaza & Salas (2021) · The macrozones locate a map maximum, not an exact epicentre

Generative exploration of aftershock sequences



Same mainshock conditions, different noise: different possible sequences

14 days · 90-minute intervals · 100 km around the mainshock

Each interval stores a maximum magnitude, with 2.5 marking absence.

> The output is a possible sequence conditioned on a mainshock.

Jara Muñoz (2025) · Undergraduate thesis in Statistical Engineering · Universidad de Santiago de Chile

One sequence mixes absence and magnitude

Representation in the manuscript

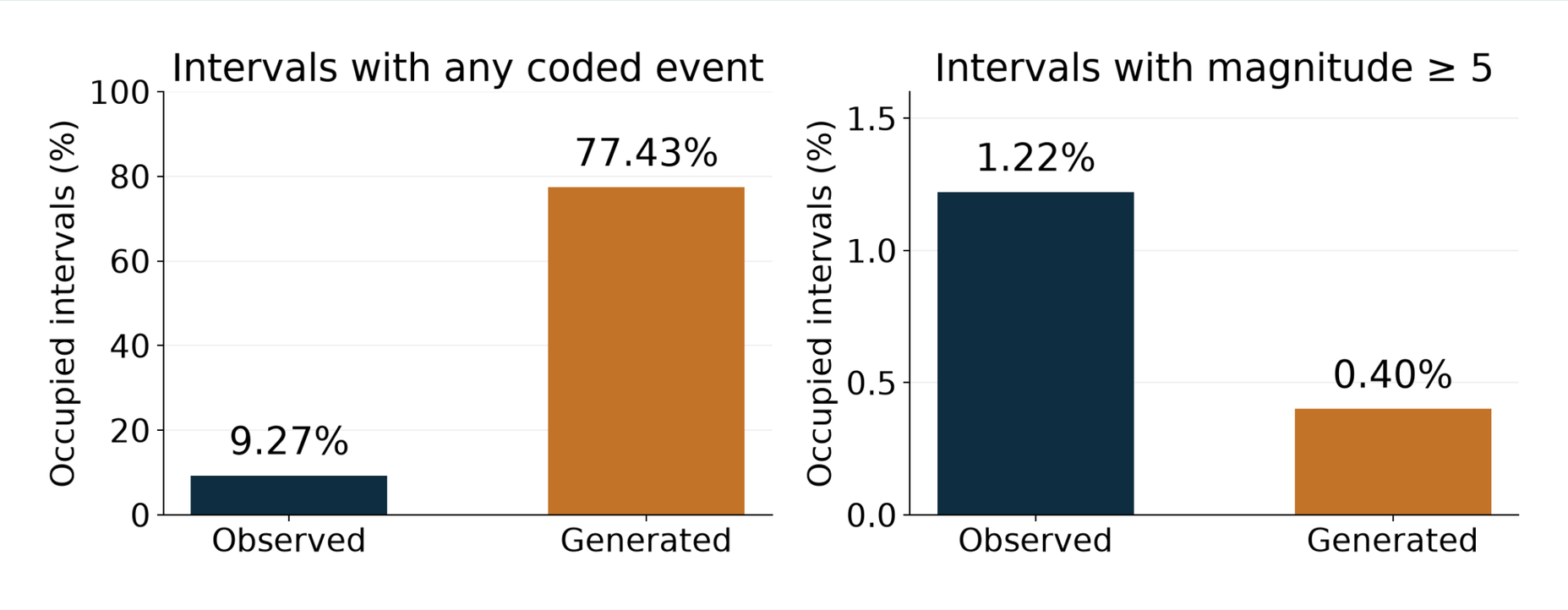
14 days → 224 intervals of 90 minutes → maximum magnitude per occupied interval



Schematic values. Gray: absence marker. Orange: maximum magnitude in an occupied interval.

> “No recorded event” and “event magnitude” are different outcomes.

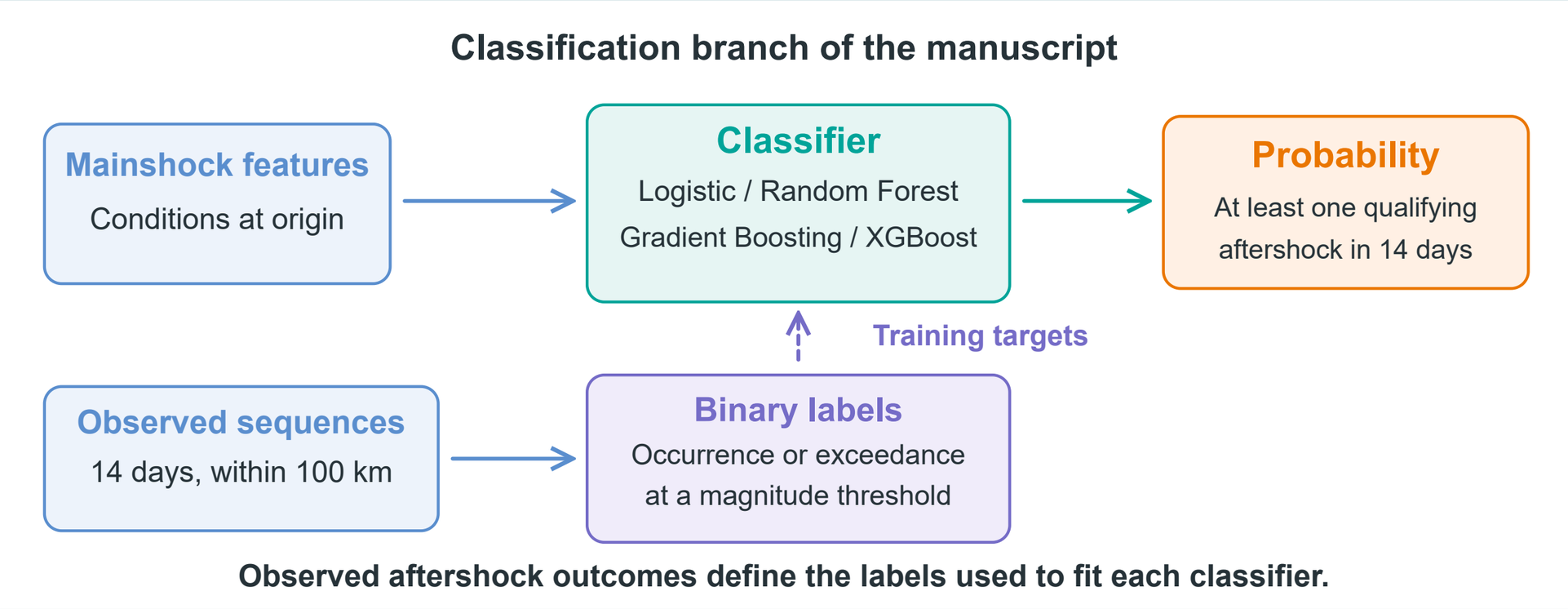
Generated sequences distort event occupancy



Assess which properties of the observed sequences are reproduced.

2026 manuscript under review · Occupied intervals, not total events · Source-case results

Observed sequences define magnitude-threshold labels



> The classification target comes from observed aftershock outcomes.

Plaza-Vega, Jara, Nicolis & Salinas (2026), manuscript under review · 14 days and 100 km

Research hypothesis: ETAS-informed generation

History available at the origin → ETAS-derived information → Conditional generator

Possible role: supply a conditioning variable or a constraint on generated sequences.

Question: does that addition improve occurrence, magnitude and temporal structure under a fixed evaluation?

> This is a proposed research direction. No hybrid results are shown here.

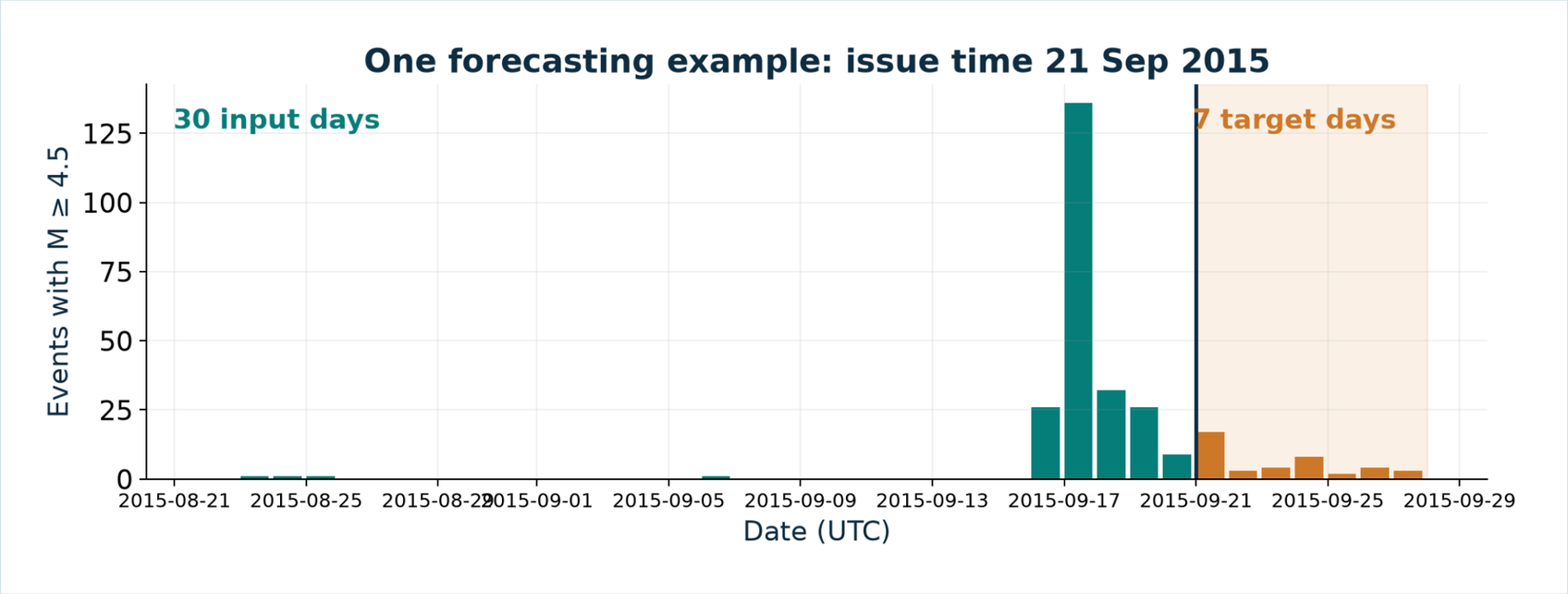
Future work discussed in the manuscript · The tutorial's illustrative activity score is a separate feature experiment

The practical applies the same modelling decisions

Case	Representation	Output
Spatial Statistics	ETAS maps and daily maxima	Intensity maximum and its macrozone
Generative study	Mainshock conditions and noise	A possible binned aftershock sequence
Classification study	Mainshock predictors; observed outcome labels	Occurrence or exceedance probability
Today's practical	Daily catalogue channels	Weekly occurrence probability

Core example: LSTM. Further comparisons: Transformer and spatial models.

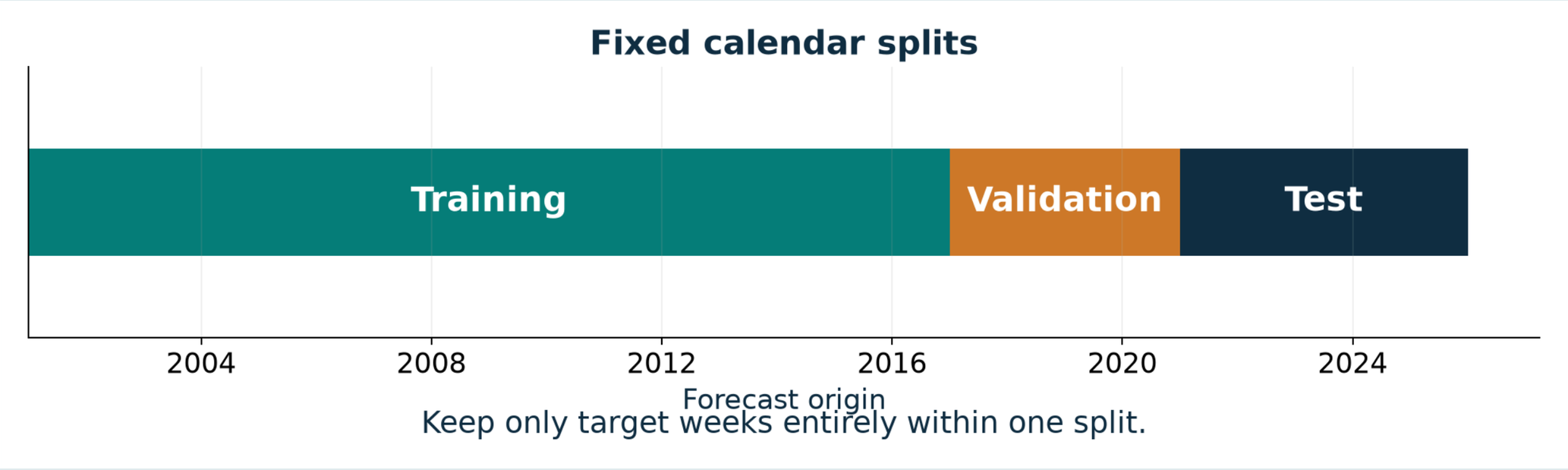
Today’s target: at least one event next week



> 30 past days → probability of at least one catalogue M ≥ 5 event in seven days.

One input: 30 days × 3 channels · Region: 28–34°S, 69–74°W · Input M ≥ 4.5 · Monday origins

Evaluate using the temporal order of the problem



Today we compare validation probabilities using Brier score, log loss and calibration.

Train 2001–2016 · Validation 2017–2020 · Test 2021–2025 was inspected previously and is excluded from this comparison.

Experiment: how does the same history enter each model?

30 days × 3 catalogue channels → LSTM → weekly probability

The same history → Transformer → the same target

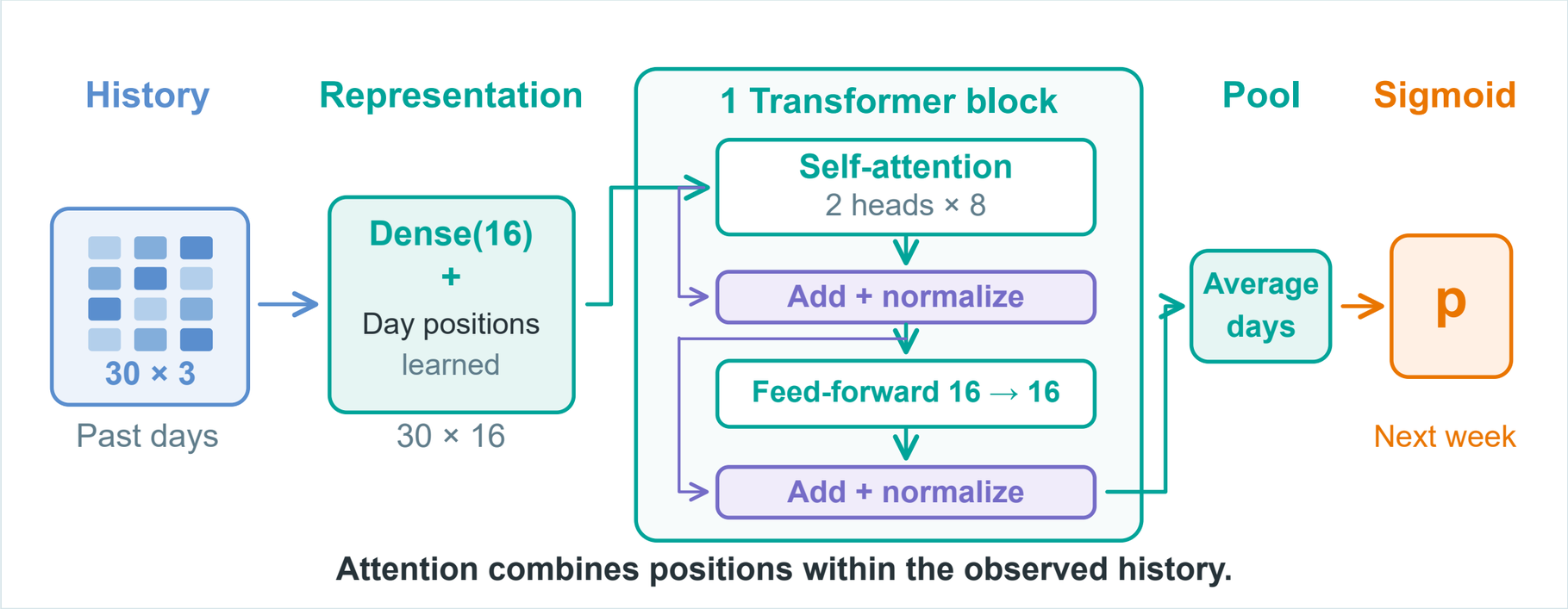
Build one example: identify the input window, its origin and its label.

Train and evaluate: read `compile`, `fit`, `predict` and compare probability errors.

> What changes when different architectures process the same information?

Temporal tutorial and notebook · Seed 7 · Optional MLP and CNN1D exploration · Activity-score experiment as an extension

A Transformer combines the same history differently



Same history and target. A different way to combine information across days.

Current temporal tutorial · 2,257 trainable parameters versus 1,297 for LSTM · One fixed, small configuration

Validation performance for the weekly occurrence task

Model	Validation Brier	Validation log loss
Training prevalence	0.187690	0.562855
Logistic regression	0.203664	0.614405
LSTM	0.181521	0.547512
Transformer	0.187901	0.562477

Lower is better. Compare these scores with the calibration plot and its sample counts.

> These are development results from one region and one validation period.

One training run per neural model, seed 7 · 208 validation weeks

What can this result tell us about the problem?

Reading the validation results

Compare each model with training prevalence.

Read Brier, log loss and calibration together.

Questions for the next experiment

Which representation and target preserve useful structure?

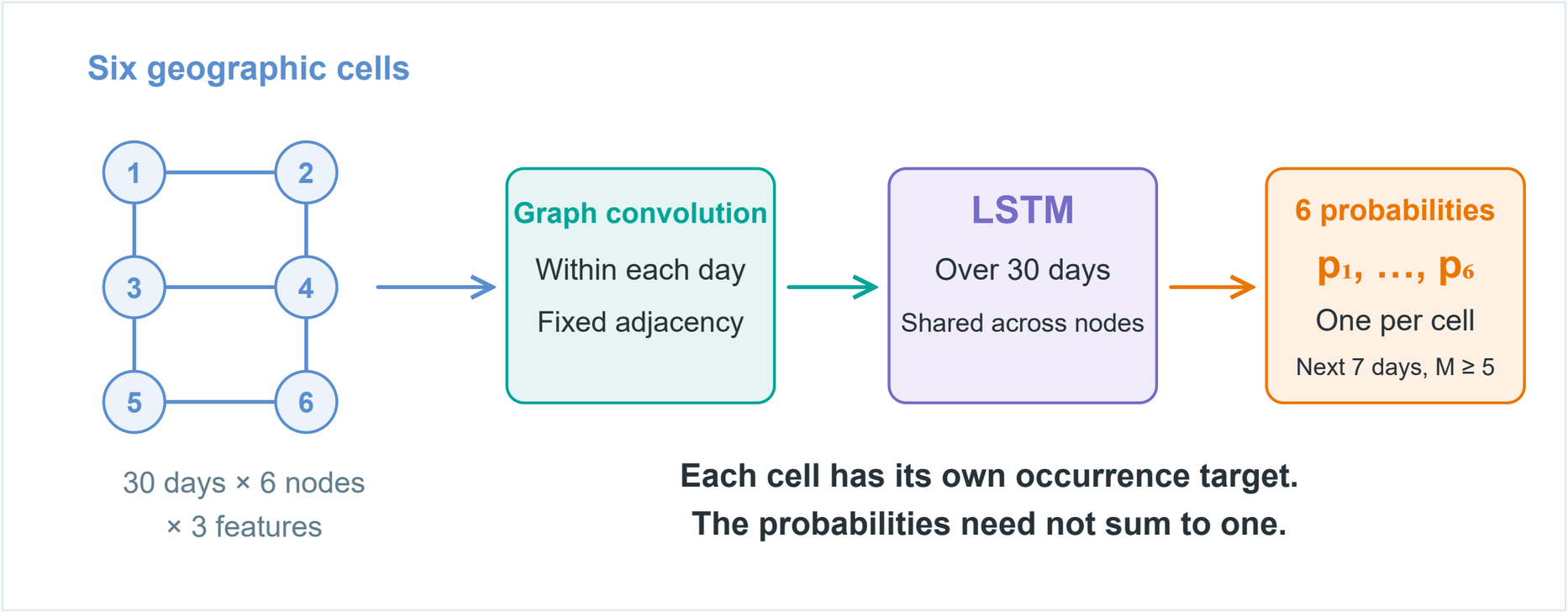
What information is available in the catalogue?

Would more data, capacity or a different training budget change the result?

> Useful deep learning depends on how we represent the phenomenon and formulate the task.

Validation also controls early stopping. One seed does not measure initialization sensitivity.

Spatial extension: Graph + LSTM



Graph connections share spatial information; the LSTM summarizes each node's history.

Input/output introduction · Complete graph example for optional exploration

What did we learn, and what would we test next?

1. What information did our daily channels preserve or discard?
2. What did the LSTM and Transformer comparison establish?
3. Which research question would you explore next with maps, graphs or an activity score?

> Deep learning offers flexible tools. Their scientific value depends on the question and the evidence.

Further exploration: optional reading and experiments · LSTM and Transformer resources

Appendix · the LSTM steps

```
inputs = keras.Input(shape=(30, 3))
state = keras.layers.LSTM(16)(inputs)
probability = keras.layers.Dense(1, activation="sigmoid")(state)
model_lstm = keras.Model(inputs, probability)
model_lstm.compile(optimizer="adam", loss="binary_crossentropy")
history_lstm = model_lstm.fit(
    X_train, y_train, validation_data=(X_val, y_val),
    epochs=20, batch_size=64, callbacks=[stopping_lstm], shuffle=False)
p_lstm = model_lstm.predict(X_val).ravel()
brier = np.mean((p_lstm - y_val) ** 2)
```

> Define the model · compile · fit · predict · evaluate probabilities

Appendix · quantities and scientific meaning

Quantity	Question answered
Process intensity	At what conditional rate do events occur?
Magnitude	How large is an event?
Shaking intensity	What motion or effects occur at a location?
Risk	What consequences may occur given exposure and vulnerability?

> Today’s output is a probability of a defined occurrence event.

Appendix · ETAS-inspired feature experiment

$$r_t(s) = \sum_{t-L \le t_i < s} 10^{\alpha(M_i - M_0)} \left(1 + \frac{s - t_i}{c}\right)^{-p}, \quad s \le t.$$

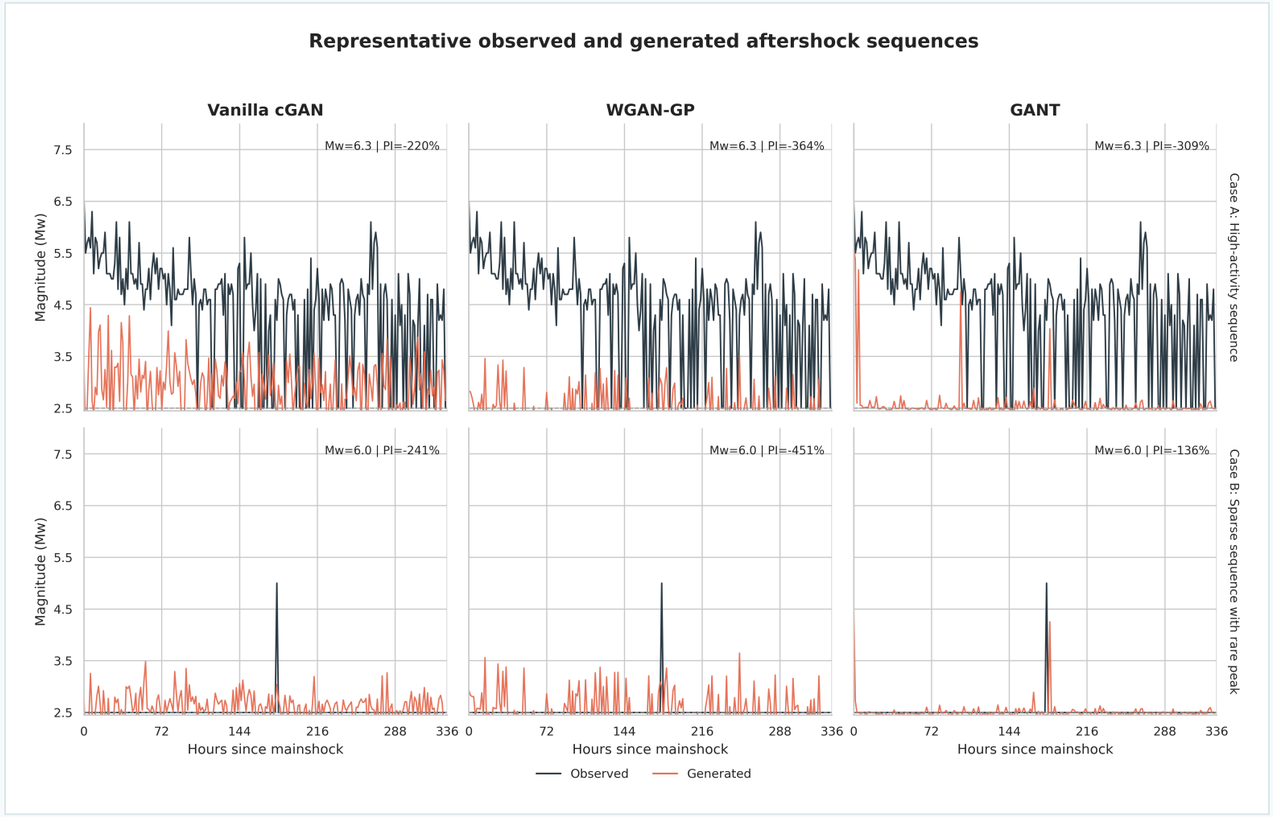
Illustrative: α=1, c=1 day, p=1.1, L=30 days.

Controlled comparison: both LSTMs use the same available history.

Representation: log1p(score) is an additional daily channel.

> Optional representation experiment. The illustrative index is not a fitted ETAS rate.

Appendix · original generative examples



2026 manuscript under review · Original PI annotations use a mean-reference denominator · Illustrative cases

Appendix · evidence and availability

Check	Meaning
Chronological, purged targets	Future target intervals do not cross partitions.
Training-only normalization	Later inputs do not set scaling parameters.
Frozen revised catalogue	Data bytes and provenance are reproducible.
Historical catalogue vintages	**Not reconstructed.**

> Reproducibility and historical information availability are separate requirements.

Appendix · sources and further reading

- **Seismic applications:** ConvNetQuake · RECAST.
- **Our published case:** Nicolis, Plaza & Salas (2021).
- **Aftershock studies:** Jara Muñoz (2025), Statistical Engineering undergraduate thesis, Universidad de Santiago de Chile; Plaza-Vega, Jara, Nicolis & Salinas (2026), manuscript under review.
- **Attention context and further reading:** Delgado et al. (2025) · Farfán et al. (2025).
- **Diagram style:** author's DL course, Units 2–5 · Transformer schematic matches the current temporal tutorial.
- **Materials:** practical · Further exploration · full references.